MILITARY SPECIFICATION
RESOLVER, ELECTRICAL, AC
SPECIAL MOUNTING FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
TYPE 54RU10N8

1. SCOPE

1.1 This specification establishes the requirements for a Resolver, AC, Type 54RU10N8, Special Mounting, hereinafter termed resolver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-4456	Shock, Variable Duration, Method and Apparatus for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-R-21530	Resolvers, Electrical, AC, General Specification for

DRAWINGS

Bureau of Naval Weapons

2104197	Resolver, Azimuth Axis

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. - The resolver specified herein shall be in accordance with MIL-R-21530 with the exception of the following paragraphs:

3.4.24, 4.7.4.3.5	Brush slip ring continuity
3.5.1	Vibration, operating condition
3.5.3	Shock, operating condition
3.5.5, 4.8.6	Tropical fungus environment
3.5.6, 4.8.5	Salt water (corrosion)
3.5.8, 4.8.11, 4.8.12, 4.8.13	Operating temperature

3.1.1 Insulation resistance. - Paragraph 3.4.9 of MIL-R-21530 shall apply with the exception that the resistance values specified in 3.3 herein shall apply.

3.1.2 The resolver specified herein shall be tested as an RX only.

3.2 The resolver shall be of the design, construction, and physical dimensions as specified on Dwg. 2104197 and as specified herein. The design and construction of the resolver shall not be changed after acceptance and approval of the preproduction sample or of the first production lot without prior approval of the procuring activity.

3.3 Specific requirements. - The resolver shall have specific characteristic requirements as follows:

Primary winding	Stator
Rated primary voltage	115 volts root mean square (v rms) ± 3 percent
Test voltage (primary)	115 v rms ± 2 percent
Test frequency	800 cycles per second (cps) ± 0.5 percent
Nominal input frequency	800 cps
Temperature rise	20 degrees centigrade (C) max. at 25 degrees C ambient
Friction torque	10 inch ounces max. with applied axial load of 175 + 25 - zero lbs

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Weight	1000 ± 50 grams
Primary current	113 milliamps nominal
Primary power	2.3 watts nominal
Stator impedance, Zso	180 ± 40 percent ± j 1000 ± 10 percent
Rotor impedance, Zrs	50 ohms minimum, 100 ohms maximum, complex impedance
Impedance balance	10 Z-sec. max.
Transformation ratio (Ke) (see note 1)	1.000 ± 0.5 percent
Phase shift, stator to rotor (see note 1)	1.25 ± 0.5 degrees
Phase shift vs rotation	± 0.5 degrees max.
Phase shift vs voltage	± 0.5 degrees max.
Primary tuned open circuit impedance	4000 ohms minimum
Total null voltage	60 millivolts rms max. at 22 ± 2 degrees C
Fundamental frequency null voltage	40 mv rms max. at 22 ± 2 degrees C
Angular error (see note 2)	20 seconds of arc max. at 22° ± 2 degrees C
Maximum rms error from electrical zero	10 seconds of arc
Null shift/par. 4.2 herein	20 sec (max)
Angular intervals for measurement of accuracy	5 degrees
Angular error versus voltage	5 seconds max.
Angular error versus frequency	5 seconds max.
Angular error versus vibration	10 seconds max.
Angular error versus humidity	5 seconds max.

Note:

1. Transformation ratio and phase shift shall be measured with a load of 1900 ± 1 percent + j 4323 ± 1 percent. The inductive component of this load shall be an 860 MH choke, cat. MP-930-30, manufactured by Communication Accessories Company, or equivalent.

2. Angular error requirements shall be met under the following conditions of mechanical loads and misalignment:

   1. 175 + 25, -0 pounds axial load and either a moment of 200 ± 20 inch pounds applied to the rotor about a point on the axis of rotation and at the center of the bearing pair, or with the resolver mounting surfaces skewed by an angle of at least one

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milliradian. The moment or skew shall be applied to the rotor at the mechanical angle corresponding to electrical zero.

2. 150 ± 20 pound radial load.

Insulation resistance (minimum)

Windings to case ..... 100 megohms
Between Windings ..... 10 megohms

3.4 Environmental requirements

3.4.1 Temperature, operating.- Ambient air temperature between the limits of + 57 ± 5 degrees C, unless otherwise specified.

3.4.2 Vibration, operating.- The resolver shall be capable of withstanding the vibrational forces specified in table I in each of three mutually perpendicular planes, at 22 ± 3 degrees C, in accordance with paragraph 4.8.2 of MIL-R-21530.

Table I. Vibration During Operation

Frequency (cps)	Acceleration amplitude	Frequency sweep rate	Test Duration
5 to 15	0.030 ±0.006 inch peak to peak	Per MIL-STD-167, type I	Per MIL-STD-167 type I
16 to 25	0.020 ±0.004 inch peak to peak	Per MIL-STD-167, type I	Per MIL-STD-167 type I
30 to 2000 sinusoidal	5 grms ± 3 db	Logarithmic; from 20 to 2000 in two minutes	Two minutes
100 to 200 sinusoidal	10 grms ± 3 db	Logarithmic;from 100 to 200 in 20 seconds	20 seconds

3.4.3 Shock, operating.- The resolver shall be capable of withstanding the following shocks in three mutually perpendicular planes at 22 ± 3 degrees C, in accordance with paragraph 4.8.9 of MIL-R-21530. Test method and apparatus shall be in accordance with MIL-S-4456.

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3.4.3.1 Two impact shocks in each plane of 35 g ± 10 percent peak amplitude and 11 milliseconds ± 10 percent duration.

3.4.3.2 Two impact shocks in each plane of 50 g ± 10 percent peak amplitude and 3 milliseconds ± 10 percent duration.

3.4.4 Acceleration, operating.- The resolver shall be capable of withstanding acceleration in each of three mutually perpendicular directions consisting of 20 g maximum when tested at 22 ± 3 degrees C, in accordance with procedure I of MIL-E-5272. The resolver shall show no damage after acceleration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance.- Sampling and inspection shall be in accordance with MIL-R-21530.

4.2 Operating temperature test.- Preproduction and periodic production samples shall be tested as follows:

4.2.1 Maintain the resolver ambient temperature at 22 ± 2 degrees C.

4.2.2 Mount the resolver on the angular accuracy index stand and apply excitation voltage to the S_1-S_3 windings.

4.2.3 Check angular accuracy as described in MIL-R-21530.

4.2.4 Set the resolver to electrical zero.

4.2.5 Increase the resolver ambient temperature to 57 ± 5 degrees C and hold for a period of 2 hours.

4.2.5.1 Check the angular shift of the electrical null point by means of a mirror and autocollimator. This shift shall be less than 20 seconds of arc.

4.2.5.2 Again check the angular accuracy of the resolver. The change in angular accuracy shall be less than 5 seconds of arc.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-21530.

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6. NOTES

6.1 The notes of MIL-R-21530 are applicable herein.

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